

DUNE's DAQ architecture; Envisaging the alternatives *

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Abstract

This document reviews the alternatives for the DUNE TPC DAQ hardware platform in a general way. We first describe the requirements as far as they are known at present and the detector layout (briefly). To avoid giving a long list of strategies, we give two example architectures which illustrate the general selection of strategies available to us (two which are slightly different to the ones proposed by other groups). Then we discuss and compare the advantages and disadvantages between all these schemes and others that have been proposed. This first version of the document is intended partly to open up the issues for discussion and also to identify some of the R&D that could be done to get started.

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*This is a draft internal note for Daq group, so some parts has been removed for public release

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1 Introduction

The Deep Underground Neutrino Experiment (DUNE) project aims to be operational in 2026. The current far detector design is for four modules of instrumented liquid argon (10 kt fiducial each); the first two modules are expected to be completed in 2024, with the beam operational in 2026 and the final module operational in 2027. This document's focus is on the hardware platform for the single phase TPC DAQ readout. The photon detector, DAQ software and back-end DAQ will not be discussed. Each module comprises of 150 Anode Plane Assemblies (APAs), the current configuration has 2560 wires per APA. About one-third of the wires are collection-view wires with unipolar signals. The other wires are induction-plane wires with bipolar shaped signals and are somewhat smaller. The TPCs are sampled at the rate of 12 bits@2.0 MHz and the readout window per event will be somewhere in the range 3.0 to 5.5 ms (for a 2.25ms drift time). The operational lifetime expectation of DUNE is at least 20 years. Working back from the aim of having the first 10kt ready by 2024 (150 APA). This implies, assuming a production rate of one DAQ unit (per APA) in two days, the starting date for production should be 2021, to leave a year for assembling and commissioning. This timescale needs DAQ R&D to be finished by 2020 to leave a year for final technical and budget approval and tendering. This suggests 2017 is an ideal starting time for R&D. (Table 1). This is an initial estimation timeline to indicate when we have to begin the R&D program)[1].

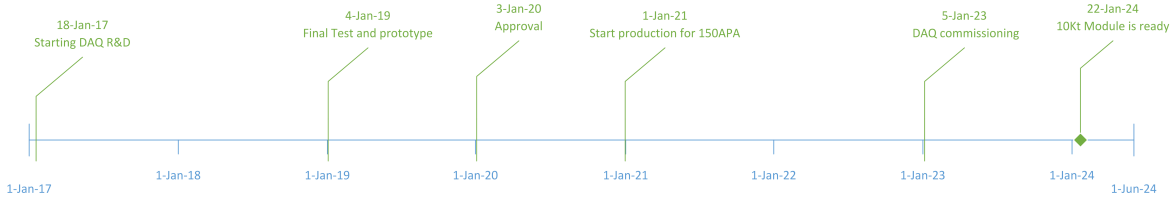


Table 1: DUNE project time line

The development work for ProtoDUNE and lessons learned from the 35t, have provided a primary outline of the physics requirements including the trigger, clock and timing distribution as well data compression, hit finding for triggering, noise filtering and other algorithms that need clearly defined hardware resources. The physics and data flow requirements, which shape the DAQ architecture directly, require study that is not included in this document but we have mentioned some of the requirements that have a high impact on the architecture design in section 1.2 . In addition, for a comprehensive study of DUNE's DAQ electronics we might also need to consider the following factors (which are more on the engineering side):

1. Hardware and maintenance **cost per channel**.
2. **The technology life-cycle (TLC)**, Components with long life-cycle to mitigate the risk of obsolescence of essential components including FPGA, network components. The best scenario is to design an architecture using a maximum of components from off-shelf technologies with guaranteed long TLC and availability.
3. **Component power consumption**; At the detector side, since it is deep underground, power consumption will increase the cooling infrastructure and running cost.
4. Maintenance, reliability and lifetime; **The DUNE project is expected to run for 20 years**. So **lifetime and reliability** of components and design have higher priority relative to other projects. Note that a design with more of the components on the surface will likely have easier maintenance, this is highlighted in the deigns we have considered (section 3). The life time and reliability (guidelines) will be discussed in section 2.1.3 and further in detail in a later document.

5. **Emphasize modular design**; of both hardware and firmware to avoid a change affecting the rest of the sub-modules, all blocks specs and interfaces need to be well defined
6. **Rapid R&D and firmware modification**; The hardware and tools used in the design will affect the complexity and time needed for further development. As the current users of the DAQ are mostly physicists, using the tools that have more layers of abstraction and similarity to C++ could reduce a notable amount of time needed for modifying, adding new algorithms and so on by users.

We need to estimate the data-transfer, buffering and processing capabilities needed in the different parts of the system, however, currently, there remains some ambiguity in the required detector performance, because the exact characteristics of the noise in the far detector are not known. The work on detectors like ProtoDUNE, 35t, ICARUS and MicroBooNE have laid down important studies, many of which are ongoing.

Our suggested strategy for this is (a) to specify for the electronics and detector construction a noise level that is as small as possible; then (b) to design the DAQ that will still function with a far higher amount of noise and also to function with a lower amount of noise with obvious savings in the amount of equipment that needs to be purchased, powered, cooled and maintained. Ideally the decisions of whether to apply the savings could be made after the noise levels are known. This is a tough strategy to implement, because the noise levels will only be known late, but we will look out for the possibility of applying it in section 1.2 when we summarize the designs described here. The executive summary is that to accomplish this, the hardware needs to (a) provide full-bandwidth (i.e. 8GB/s/APA) capability to transfer the data into computer memories and (b) not miss the opportunity to generate simple signal-above-threshold trigger hit finding that is optimized for the low-noise situation in hardware. This is called the high/low noise strategy in later parts of the text.

1.1 TPC Front-end Electronics and data rates

This section gives some notes on the constraints on the DAQ from the interface with the cold-electronics. The 2,560 channels from each APA are read out by 20 Front-End Motherboards (FEMBs), each providing digitized wire read-out of 128 channels. The 128-channel cards planned for DUNE will use eight 16-channel FE, eight 16-channel ADC, and two COLDATA ASICs.[2] Each ADC will transmit two streams of data at 192 Mbps for a total data input of 3.072 Gbps. All data will be transmitted off-board to the DAQ. Twelve bits of ADC data per APA channel every 500 ns yields a single-channel bit rate of 0.024 Gbps. With 8B10B encoding, for example, this increases to 0.03 Gbps, plus some overhead for frame data to indicate event blocks. Assuming a conservative serial-link transmission speed of 1.2 Gbps, a single link can therefore handle 32 channels. Thus, it is planned to drive four 1.2 Gbps links from each COLDATA ASIC.[3] (For ProtoDUNE, an FPGA mezzanine will replace the COLDATA mezzanine board which is still under development). Data will be driven to the CE flange through copper cable utilizing low-voltage differential signaling (LVDS). All data from the FEMB inside the cryostat travel through cold cable bundles to the warm outside via a flange. The data rate per APA, using the full event-buffer scheme is shown in table 2. This table is based on the current COLD electronic design however if the design changes, we still expect similar I/O rate and configuration.

1.2 Triggering

The detail of related timing&clock distribution, control and monitoring architecture are not coming in this document the major constraint on hardware system design might be forced by the

- supernova event detection that requires large ring buffer for raw data before firing the supernova trigger.

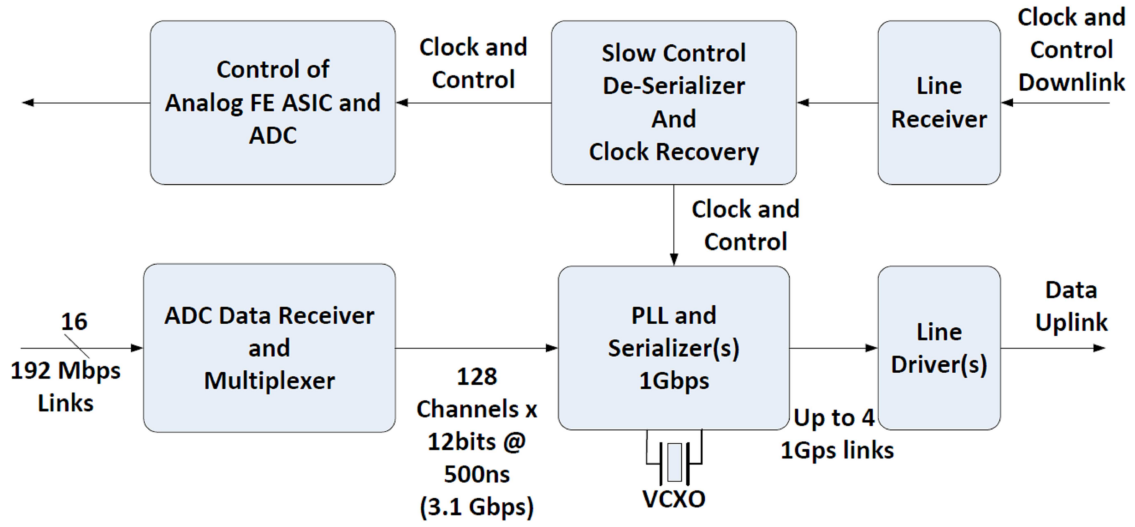


Figure 1: Functional Block Diagram of the Cold Digital Data (COLDDATA) ASIC[3]

Element	Quantity	Rate	Channels
APA	1	100Gb	2560
Front-End Mother Board (FEMB)	20 / APA	5Gb	128
ColdData chip	2 / FEMB	4×1.25Gb	128
Cold cable bundles	80/APA	1.25Gb	32

Table 2: Rates full event-buffer scheme

136 • trigger system in case triggers are required needing significant processing and bandwidth for
137 transfer to the back-end DAQ.

138 There are several idea and algorithms around, e.g. the implementation of a primitive trigger stream in
139 the front-end DAQ with the help of ring buffers, could reduce the system complexity and bandwidth.
140 Some detail of trigger hardware and data flow are in the appendixB.

141 1.2.1 Supernova trigger

142 Ideas are forming for ways of collecting supernova events, either (a) to collect the individual events
143 throughout the burst efficiently (and some randoms for efficiency studies), or (b) to readout entire
144 10+ second long windows of raw data. Note that in some cases, the supernova may not produce
145 large numbers of events near the start of the burst, but measurement at that time is still extremely
146 important, so some form of ring buffer of a few seconds to allow for trigger latencies will likely be
147 required also. To achieve these goals, the DAQ system needs either a large data buffer after the FEMB
148 or high compression and zero suppression ratio. The unknown level of noise and TPC performance as
149 well as the desirability to avoid zero suppression makes the memory buffer presence in the DAQ more
150 desirable. The amount of memory buffer needed for each 1.25Gb/s link for 10s is 1.5GB and for an
151 APA at least 124GB.

152 1.2.2 Atmospheric neutrinos, Proton Decays, Cosmic rays

153 Atmospheric neutrino events are asynchronous with any external warning that can be used as a trigger.
154 Therefore the trigger should be actively looking at all the data to find these events. Proton decay
155 candidates have this same feature. The same for cosmic ray muons (although these may hit a larger
156 number of wires). The same strategy can also be used to collect beam neutrino interactions, although
157 in that case a warning of the time of arrival from Fermilab will also be available.

158 1.2.3 The Beam trigger

159 coming later

160 1.3 Data and signal (simplified) flow

161 The primary objectives of a DAQ design are the reliable and efficient transport of data from TPCs to
162 back-end DAQ (permanent storage), accurate distribution of timing and trigger signals and dependable
163 triggering as discussed above. The secondary objective of the DAQ is continuous monitoring of the
164 data taking and the quality of data, by providing a coherent control and configuration of the front-end
165 electronics (COLDData and digitizer chips) for the data taking. The simplified and abstract data flow
166 from one APA is shown in figure 2. The Control flow and timing, trigger, back-pressure distribution
167 is shown in figure 11.

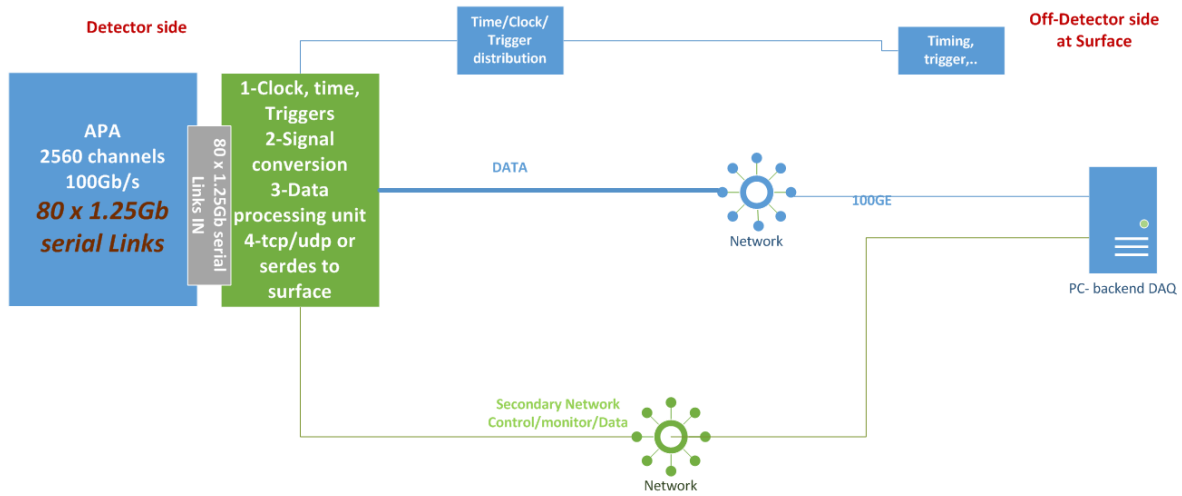


Figure 2: Simplified Signal and data flow for APA

We emphasize that the cold cable bundle with 1.25Gb serial data, the ring buffer as required for supernova events and processes such as time tagging, trigger primitives are the major issues for the designs since the detector performance (noise) is not clear yet.

2 Review current and future technology

2.1 General conceptual design requirements

The event triggering will require a continuous search and processing of the raw data stream coming from the TPC detector all the time. This can be done in two ways;

- Transferring raw data from the TPC to back-end DAQ computers and do the processing in the back-end DAQ computers or on attached GPU or FPGA boards. The rates of the full event-buffer scheme shown in table 2 require a bandwidth of 100Gb/s per APA to transfer the whole data stream to the back-end DAQ.
- Implementing trigger primitive finding in the front-end DAQ and saving the raw data locally in a ring buffer. The trigger primitives are collected and processed in a trigger farm to create a trigger decision that signals to the front-end DAQ to send contents of the ring buffer. This scheme will reduce the bandwidth needed.

The other factor here is the detector performance, in particular, the ratio of signal to noise and level of coherent noise in the TPC. The detector performance will affect the feasibility of trigger primitive fake rate and efficiency, data compression etc. We have started some studies and initially it seems the following are requirements for the architectural design of the DUNE DAQ.

1. Bandwidth needed or provided for the data stream
2. Supernova detection, trigger buffer, (atmospheric neutrinos and proton decays etc. are covered by the same requirement as the supernova expect it does not need so much buffering)
3. Detector performance, electronics' noise, level and source.

There are strong trade-offs between the above three requirements. For example, the high bandwidth networking is expensive, it could be avoided by using trigger primitives but on the other hand trigger primitive efficiency depends on the detector noise performance. One can reduce the noise in the acquired signal by applying a digital filter in the front-end DAQ before the trigger primitive generating algorithm but at the cost of using lots of gates and resources in the FPGA. Conceptual designs will be needed, considering these factors with some input from networking and FPGA selection to clarify the feasibility of each alternative architecture. Note also that FPGA pricing is not linear with the complexity on the chips; the small ones are cheap to attract mass-market customers, while the bigger ones and/or the ones with more fast serializes on them for network connections are newer and the manufacturer needs to cover design costs, so are much more expensive.

2.1.1 Networking

One of the essential parts of the DAQ chain is the networking components. This could change the whole DAQ layout significantly. The 10GE is currently the dominant technology and will remain in the market for the next decade. Almost all high-end servers have come with 10GE ports as standard networking speed since about 2016. The 10GbE managed switches will be prominent in the market and their price may fall below 2k\$ rapidly in the next few years. The high demand for higher rate pushes technology for the next 5 years to establish 100GE to 400GE as the next dominant standard technology. Regarding the IEEE 802.3cd task force, the 10GE optical networks are going to be replaced with 50 Gb/s, 100 Gb/s, and 200 Gb/s technologies in the not too distant future. [4]

Already Xilinx has included the 100GE MAC IP in the Ultrascale plus family. Using 100Gb/s on the DUNE time scale for handling the data transfer to the surface, is an efficient and feasible option that reduces network and hardware nodes. It should be straightforward to use it, by aggregating data in commercial multiplexers or switches, and can even consider using it on custom designs, since the PCB layout design and materials are almost the same as 10GbE and reduces the cabling's cost a factor of 10 times less. However on our PCB designs, it is likely that one PCB will serve a certain maximum number of channels, and the maximum data rate from those channels will only be 10GB/s or 20GB/s at most.

2.1.2 FPGA electronics

The current major FPGA vendors are Xilinx and Altera; together they control almost 80 percent of the market (a third one, Microsemi, might also be interesting for the DUNE DAQ design). Both Xilinx and Altera provide proprietary Windows and Linux design software, Vivado and Quartus respectively, which helps engineers to design, analyze, simulate and synthesize (build) their designs.

The current breakthrough in FPGA electronics is based on embedding hard-core components in an FPGA fabric to add more resources and functions (for Xilinx, this is with almost same price of the FPGA at the same class). In particular, Xilinx have the Zynq line of FPGAs, which is becoming one of their mainstream products, that include on-chip multi-core ARM CPUs, Ethernet interfaces and PCIe interfaces. Since ALTERA have recently merged with INTEL, similar things from them are likely soon.

Some Xilinx parts have had more than a twenty year life availability. The shortest life of a product was often due to it being replaced with a far superior technology or version. In any event, if one signs up for notices, we will have up to a two year warning. As Xilinx parts are used extensively in the communications and internet backbone, their customers require a 15 year operational lifetime. The Zynq UltraScale family is intended to be in high life time cycle of 15+ years. ¹

There are new C to HDL tools, like Xilinx's HLS and Altera's OpenCL. Xilinx's new SDSOC and its mature HLS tools are interesting platforms to allow more rapid high-level programming of the computationally heavy parts of projects. Hardware requirement of FPGA

¹Needs to be officially confirmed by Xilinx

- 238 • Transceiver I/O or serdes from 1.25Gb to 25Gb
- 239 • Memory interface (DDR3/4)
- 240 • TCP/IP MAC IP cores
- 241 • Very high throughput DMA
- 242 • Hard core CPU

243 Since we have so far studied the Zynq UltraScale+ in more detail than the others, here is a summary
 244 of some of its features. It is compliant with the IEEE Std 802.3ba, the 100G Ethernet integrated
 245 blocks in the UltraScale architecture provide low latency 100Gb/s Ethernet ports with hooks for user
 246 customization, statistics gathering and support for 10 x 10.3125Gb/s (CAUI) and 4 x 25.78125Gb/s
 247 (CAUI-4) configurations. [6] The core systems can be driven directly from programmable logic (which
 248 is essential for DUNE, pushing data from the CPU is not an option that will work for us) including two
 249 40Gb/s network lines for the Gigabit Ethernet blocks, TCP/IP stack processing, DDR4 memory con-
 250 trol, and dual NVMe PCIe interfaces handling up to 32TB of cache memory.[7] Each I/O pin possesses
 251 an IOSERDES (ISERDES and OSERDES) capable of performing serial-to-parallel or parallel-to-serial
 252 conversions with programmable widths of 2, 4, or 8 bits. All Zynq UltraScale devices support dif-
 253 ferential standards beyond LVDS, including RSDS, BLVDS, differential SSTL, and differential HSTL.
 254 [6]

255 2.1.3 Life time and reliability guideline factors for DUNE

256 (It is in preparation and coming later as a separate internal note.)

257 3 Alternative designs concepts

258 We present two distinct design concepts in this section. We are not pushing one of these as a detailed
 259 design proposal; rather these are feasible designs, and some combination of them, or other ideas might
 260 end up being optimal. They are similar to designs by others, and we see some advantages in them.
 261 They should be viewed a bit as us taking a devil's advocate view in describing them. We will follow this
 262 in section 4 with a general discussion of these designs, variants on them, and comparisons to designs
 263 of others.

264 Regarding the discussion of architectural design requirements in the section 2.1, DAQ architecture
 265 functionality can be divided into two parts:

- 266 a) Front-end DAQ that comprises of electronics acquires data stream from readout-electronics from
- 267 TPC detector to prepare, process and deliver it to computation farm (back-end DAQ).
- 268 b) Back-end DAQ comprises, offline storing, reconstruction computation farm.

269 There are 4 topological architecture diagrams that cover almost all the feasible architectures would
 270 fall in one of these diagrams. All these architecture concepts is shown in figure 3. The top 3 diagrams
 271 are considering to have the interface to current cold-electronics, and the fourth diagram in the bottom
 272 if cold-electronics would be replaced by warm electronics A2D.

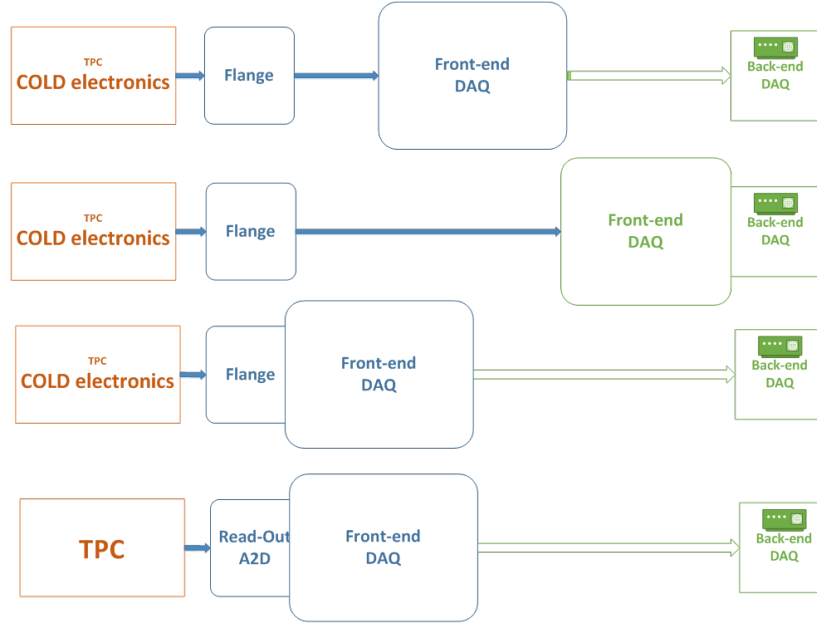


Figure 3: Architecture concepts considering the data transport and DAQ position

273

274 3.1 DaqD1: Off-detector architecture concept

275 This architecture, shown in figure 5, is based on an optical Multiplexer(MUX)/ DeMultiplexer(DeMUX)
 276 scheme on the detector side and the main processing/buffering boards in a computer-farm on the sur-
 277 face. With this scheme, we move all the data flow coming directly from the cold cables from the FEMB
 278 to off-detector via long optical fiber cable bundles. The rest of the circuits (clock distribution, control,
 279 and LV power supplies) stay like the protoDUNE design.

280 This proposed design is similar to others that have been discussed in the collaboration, for example,
 281 both the dual-phase design and the FELIX system advocate pushing all the data into the CPU farm
 282 to process there. Also, DWDM has been proposed as an extension to the WIB development at Boston
 283 University

284 This architecture is based on using Wavelength Division Multiplexing WDM of data coming from
 285 cold-cables. This could be applied also to other concepts to boost the data rate with minimum cost
 286 and complexity. There are two types of WDM architecture, Coarse Wavelength Division Multiplexing
 287 (CWDM) and Dense Wavelength Division Multiplexing (DWDM). regarding the number of cold cables,
 288 DWDM is the best fit for this concept. The figure shows a data flow and block diagram of this concept.
 289 There are two 40-channel DWDM per APA underground and two on the surface, in total four for one
 290 APA. Only one optical fibre is needed per APA, and in construction, we can use a bundle of 24 optical
 291 fibres, i.e. a total of 7 cable bundles needed for a whole detector cavern to take the full data stream
 292 to surface. At the surface in the end-DAQ PCs, the data stream would be transported to the RAM
 293 through a PCI card. In addition to the feasibility of a custom design such as a PCI card, there is wide
 294 variety of PCIe cards meeting the requirement commercially available. The design of this PCIe card
 295 is simple since it just transports the data to the RAM where further processing would be done by a
 296 GPU card. The initial estimation a 10 PCIe cards and 10 low-end GPU plugged into two servers can
 297 handle an APA for data processing, time tagging, trigger algorithm, filtering the signal and so on.

298 The benefits of this scheme are very broad from

- 299 • Significantly reduce the cost of firmware and hardware R&D (or almost no cost R&D) expect

the electrical to optical module at the APA's flanges. The DWDM multiplexers are broadly available commercially since they are used widely in the telecommunication industry. A DWDM multiplexer with 40 channels Mux/Demux + Monitor Port, LC/UPC, C21~C60 is available for less than £30 per channel.

- Highly reliable and low maintenance cost of optical components. The MTBF (mean time between failures) is ca. 500 years. The MUX/DeMUX components are passive and they need no electricity. e.g. in an underground experiment, this is a super plus.
- Almost all of the processing and algorithms needed could be handled by C++ programming and physicists.
- A high fraction of the components can be obtained and replaced independent of design during two decades of running the experiment since they are the commercial network components and PCs.
- The final comparison cost of this architecture including the total cost of R&D and manpower needed for hardware and firmware on top of the production cost of components seems promising.
- Most of the components are off-shell, which gives minimum R&D and maximum flexibility.
- It is cost effective, low maintenance and rapid R&D with minimum usage of FPGA and related custom build components.
- Commercial availability of Optical WDM Transceiver/Ethernet Optical Interfaces 100Gb/s (to 400Gb in the near future)
- Very low power consumption in the detector cavern

The above is a specific design example. More broadly, the MUX/DEMUX could be achieved in several other ways: (a) By custom designed FPGAs, this needs hardware and firmware effort or (b) Using commercial Serdes chips (needs only hardware design effort). There are wide varieties of commercial serdes (e.g. the TLK10034 [10]) that could replace FPGAs and reduce the complexity of the design but one might not achieve high bit rate and might need more optical fibre cables.

In addition to the option of using a processing farm as described and shown in figure 4, the processing could also be carried out either 1. FPGA+CPU based, in this scheme an FPGA on a PCI card installed in a server will cooperate with the CPUs to process and finally buffer the incoming data to the inboard RAM of the server. There are a large number of (cost-effective) industrial cards on the market that can be used for this propose. They will always be available in the future (long-term) in different chip manufacturers but the optical interface and I/O bus to CPU will remain the same. In the case, if technology life cycle of the card is over, this makes, firmware the only modification that would be needed if we need to use a different or new card. 2. The full FPGA based PU design. This more generic version of the design is shown in figure3.

3.2 DaqD2: RCE based architecture

In keeping with the theme of suggesting slightly devil's-advocate schemes here, we propose a variant on the RCE scheme in use in protoDUNE and the 35t. It aims to keep all the firmware features of the RCE design, but to optimize away some of the components on the COB that are not essential for the DUNE application.

The suggested architecture is to use a single carrier board (SSCB) based on the RCEs that can accommodate a common point-to-point architecture by removing the Cluster Interconnect (CI) hardware, firmware and software. The cost and time needed for the research and development of hardware and firmware, as well as the total hardware deployment and maintenance cost are always the most important factors in data acquisition systems. Our experience with using the SLAC RCE system has

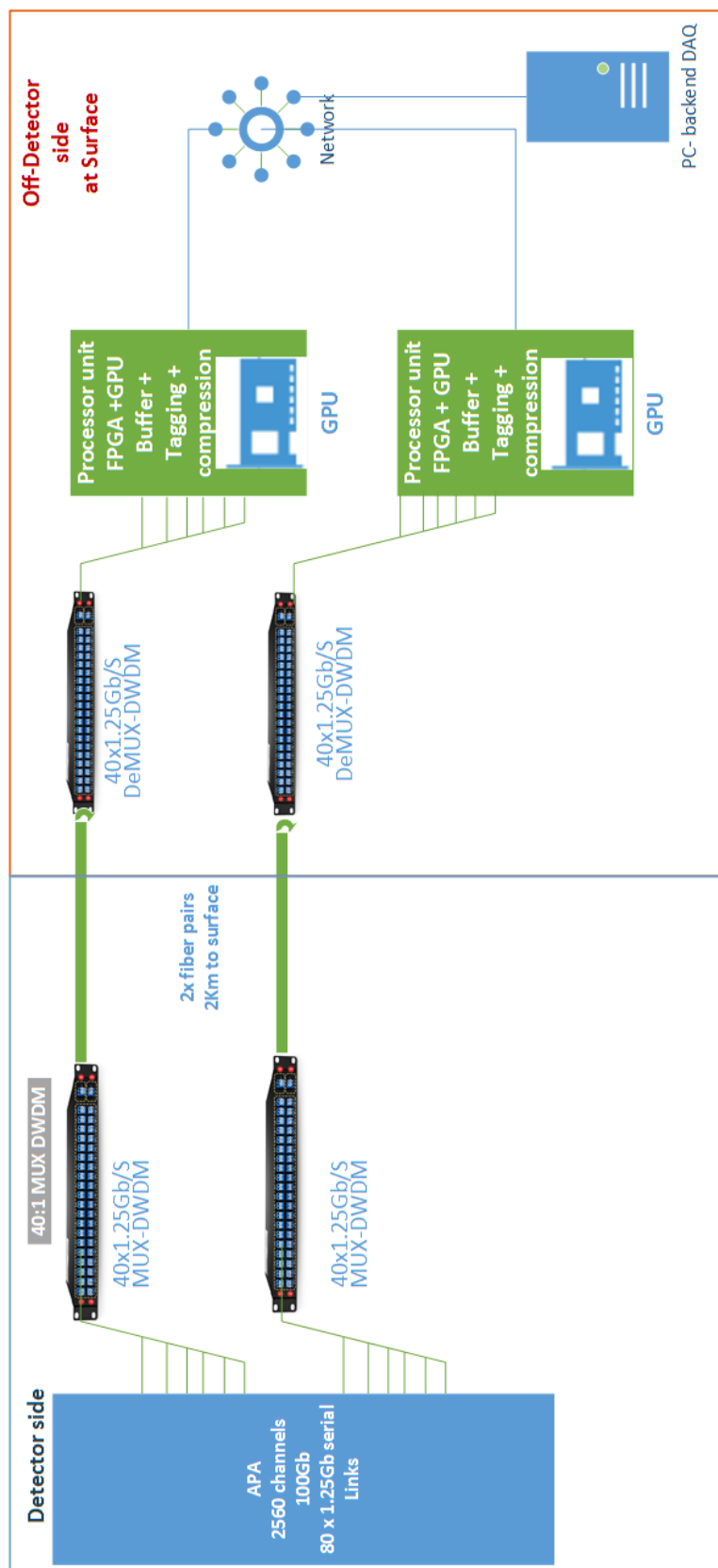


Figure 4: Off-detector architecture concept DWDM

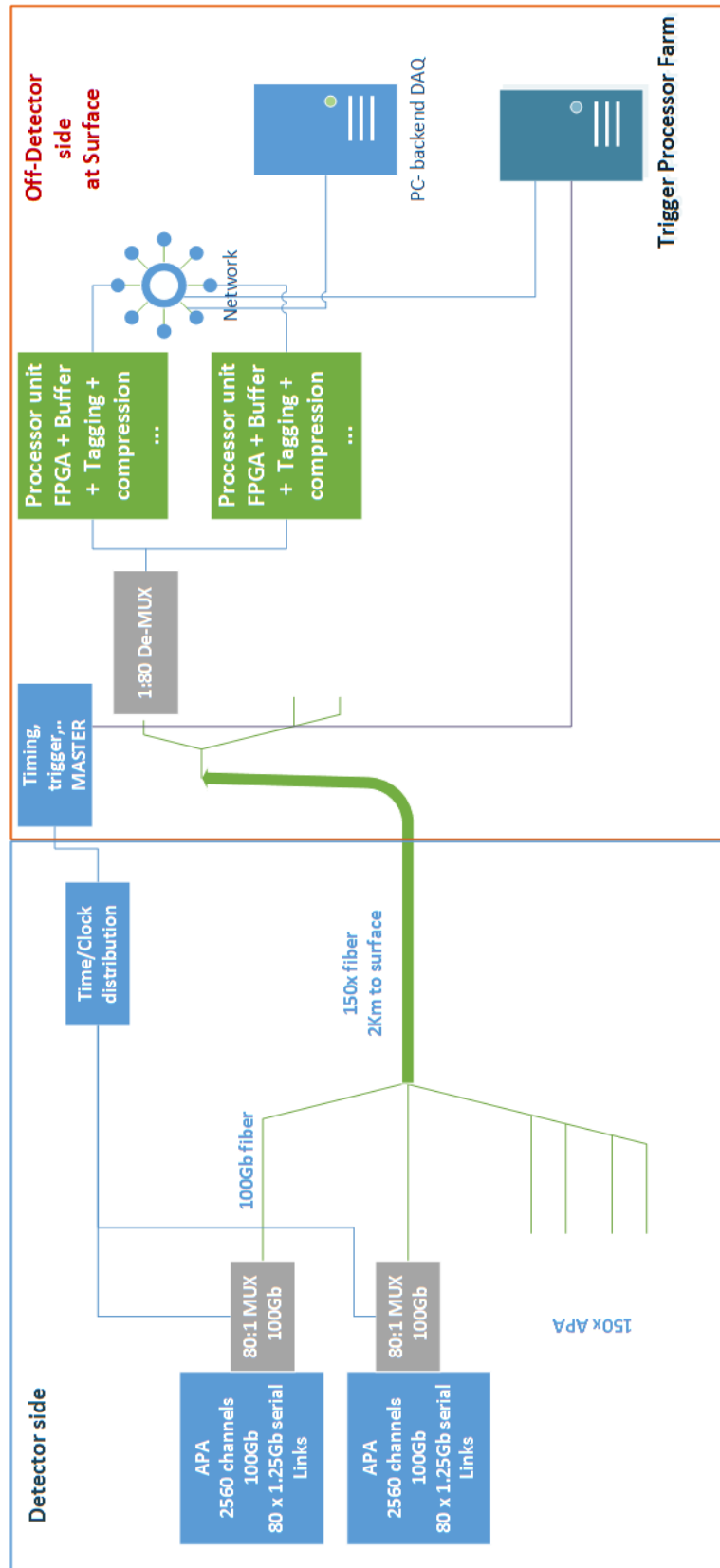


Figure 5: DaqDL, Off-detector architecture

highlighted the advantages of using a generic design that is able to be adapted to many different applications. The idea is to make a second carrier board for RCE modules that is optimized specifically to this DUNE application. Since the SSCB does not contain a CI, the ATCA could be replaced with a cheaper industrial or custom-built platform to further reduce complexity and cost. Effectively this devil's advocate suggestion is to study the optimization of the tradeoff between saving development costs (which would be absent by using the RCE/COB as-is), compared to optimizing and reducing the functionality to the minimum needed for DUNE. An advantage of the redesign is that we can also optimize the networking to allow all the data to be sent to the computers (see the high-low noise comment at the end of the introductory section 1).

The data flow in DUNE is almost always point-to-point starting from the detector front-end electronics to the back-end DAQ hardware that builds the events and saves them in storage. The algorithms for triggering can use a group of adjacent cells to deal with e.g. coherent noise, but there is no a requirement in this particular case for widespread ‘horizontal’ transfer of data from one group of channels to the next. In all the architectures considered here, such horizontal transfer that is needed in the trigger algorithms can be done further downstream in the computers.

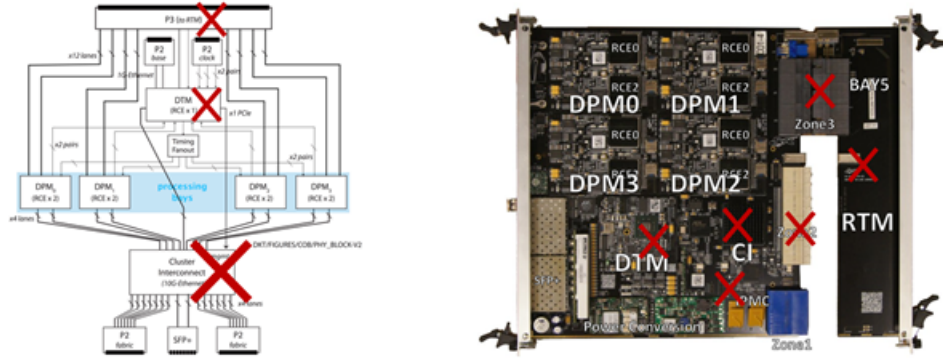


Figure 6: Indication of parts of current COB hardware that will be omitted in the SSCB board. Original photo is courtesy of SLAC

This can reduce the hardware cost for large detectors, and reduce maintenance of more complex firmware and related software. A brief overview of hardware that could possibly be removed from the current COB hardware and cluster interconnect architecture is shown in Figure 6.

The conceptual block diagram of the proposed SSCB is shown in Figure 7. The carrier board can host say 8 (or 4) RCEs modules mounted on mezzanine boards with 2x12 SFP+ serial links that stream data with maximum 10Gb/S input from the detector at one end and two 10GE output links to the other end. The aggregation that is provided by the switch of the current COB is instead provided from an external switch in this suggested system. There is an on board central controller based on a Zynq SOC that handles signals from other sources such as timing, triggers, monitoring and slow control. In addition to using a redesigned COB, this architecture will also be able to use an upgraded DPM module, to take advantage of a newer (e.g. Ultrascale+) architecture chip (see the description of DaqD4 below). By making the new DPM compatible with the old one, it will allow parallel development of the SSCB, the new DPM and the firmware, which can be tested using the existing framework from protoDUNE. A cost estimate has been started but is not included in this note yet. Some of the specifications and advantages are :

1. Adding additional on board RAM (DDR3/4) in SSCB to buffer huge data for supernova event.
2. Capability of interconnection of the links with significantly lower cost with off-shell components.
3. Capability of higher data rate throughput then current COB.

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Table 3: Preliminary Cost estimation of different design

377 4. Highly flexible and scalable architecture.

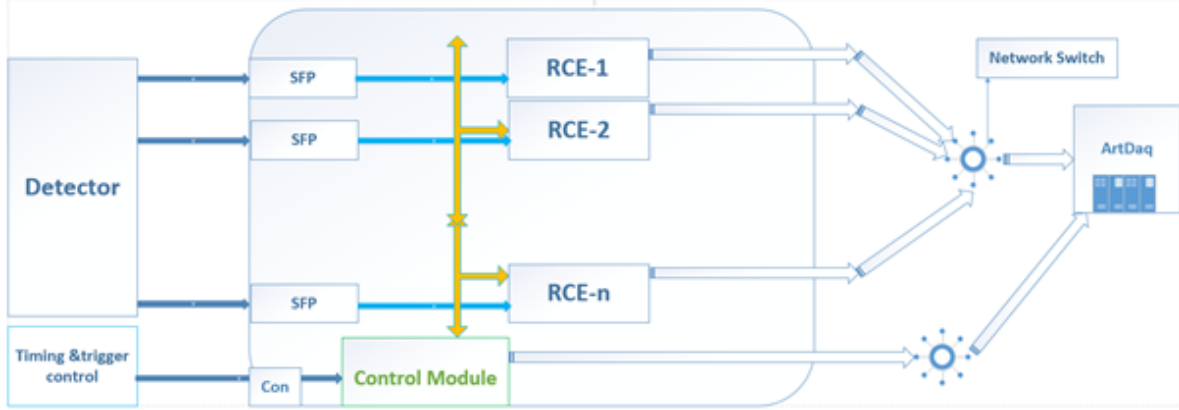


Figure 7: DaqD2, RCE based architecture

378 3.3 Other DAQ schemes (DaqD3 and DaqD4)

379 The design in section 3.2 (named DaqD2) is a somewhat customized step away from the RCEs. A
 380 further step that could be taken, but still retaining the full firmware and software compatibility with
 381 the RCE design concept is to move the DPM module instead of to a ‘lite-COB’ carrier, to move instead
 382 to the flange board itself. This has the drawback that cooling and a clean operating environment will
 383 be needed near the flange. This scheme is shown as DaqD3 in figure 8.

384

385 The DaqD2 and DaqD3 schemes can also be augmented by developing a new DPM carrier board –
 386 with similar architecture, but using the next generation of chips. This is an upgrade path that SLAC
 387 have followed for a long time, the current COB/RCE used at protoDUNE is already generation 3.
 388 We are in discussion with SLAC who are seriously considering the next upgrade path. It would be
 389 interesting if DUNE as a ‘high volume’ customer of DPM modules could indicate which is our preferred
 390 pricing point for the size of chip and amount of memory to put in the FPGA. e.g.

391 • Pricing point 1 ; A single Zynq UltraScale+ (tentatively or other FPGA with similar specifica-
 392 tions) per APA with 32GB DDR4 ram (or more on board) will drive in the 80 LVDS and send
 393 timing/clock/commands and control signals back

394 • Pricing point 2 ; Five ZYNQ UltraScale+ or similar to handle the data from an APA in five
 395 threads (or more). This allows to provide more FPGA resources by splitting i.e. it gives us access
 396 to more memory for buffering and gates for processing data as well more network bandwidth.

397 A further step which we call DaqD4 is to study a minimal FPGA scheme, in which the FPGA is
 398 squeezed to a low-cost Zynq chip servicing a certain number of optical links from the cold boards. It
 399 has just enough functionality for the input de-serialisers, a 10Gb/s output link and a ‘low noise only’
 400 trigger (see high/low noise discussion at the end of section 1.2). If the noise turns out to be high, we
 401 abandon the trigger logic here and buy sufficient computer power to derive the trigger hits from the
 402 full data stream. Note: We should ask the electronics group whether 1.25Gb/s is really their favored

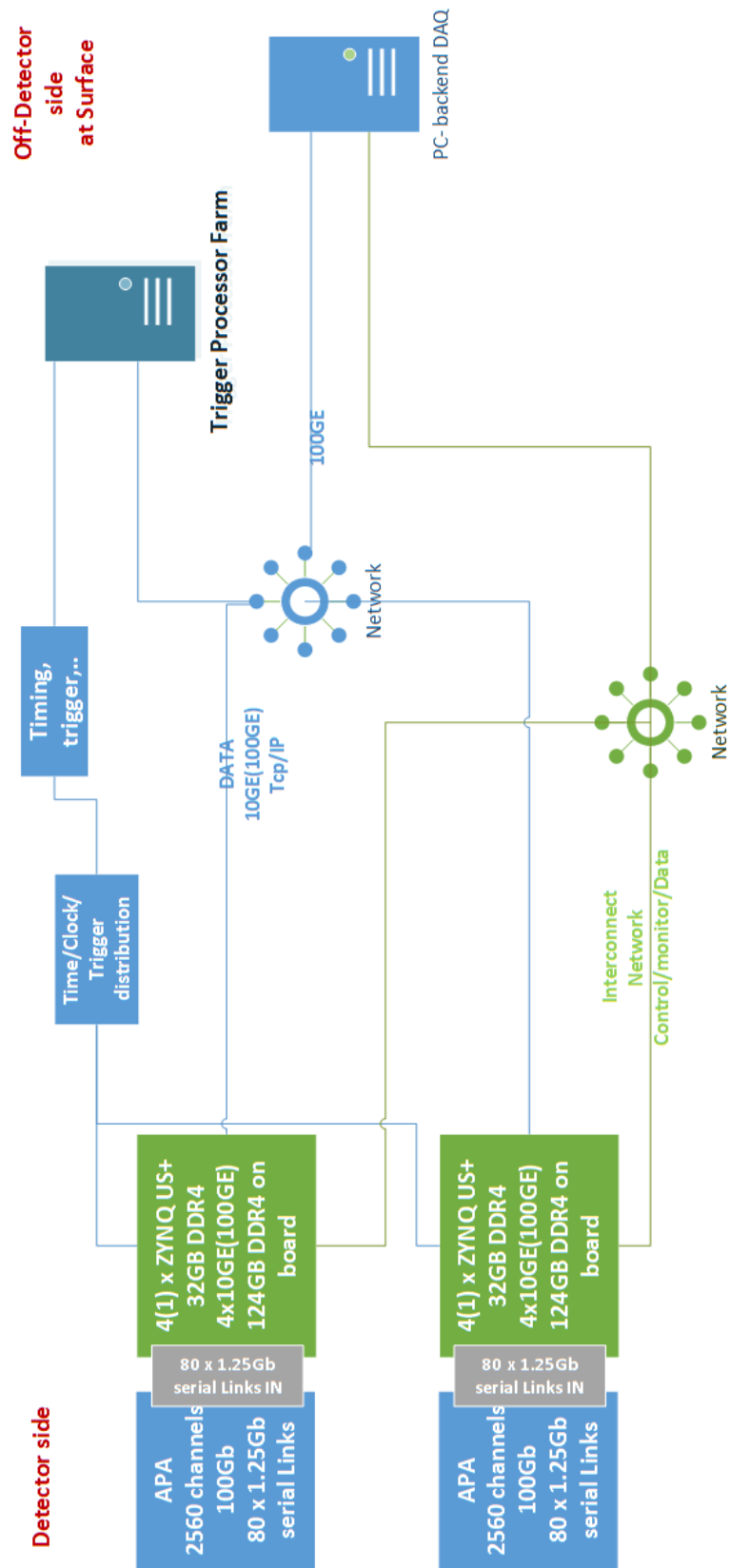


Figure 8: Flange embedded DAQ

data rate, if they reduced it by a factor of say two, it may be easier for them, would only increase the cabling by a factor two (which is not important, since the data cables no longer dominate the internal cabling, the slow controls and photon detector cables do), and would allow us to use normal pins for the input data (at 1.25Gb/s it is marginal).

The design shown in section 3.1 follows the paradigm of several other proposed solutions for DUNE of collecting the data and pushing it into computer memory. E.g. the dual-phase design and FELIX. The advantage is that all the data are in the computers to be processed should the noise levels be high. We could also consider adding a small FPGA on the input end of the links, just outside the flange, which could do ‘low-noise only’ trigger hit finding. With this small modification, DaqD1 is really rather close to DaqD4.

3.4 Comparison between Architecture concepts

The minimum major functionalities needed in front-end DAQ comprise following items (not including the trigger farm):

1. Hook to add large ring buffer e.g. 124GB per APA for supernova buffering
2. DMA to RAM for Ring buffering
3. Timing tags
4. Trigger primitive algorithms
5. Signal processing like noise filtering for trigger hit finding , data compression, ...
6. Data (buffered) transfer to back-end DAQ (either TCP/IP or custom links)
7. Monitoring ,.....

The amount of resources need to handle these functionalities could be translated to equal (at least) a Xilinx FPGA with around 4m to 7m LUT or 5 to 10 TESLA K40 GPU per APA . ²

The table 4 shows a crude comparison of the mentioned architectures. The rating is from 1 to 5, 1 is the worst and 5 is the best. Note that in the FELIX case, we have not studied the design for a long time, and there are some wild guesses here - consider this version a draft.

	Hardware cost	Filtering high-noise	Simplify low-noise	R&D	TLC	Fast night repair	scale-abiity	COS	Operating cost
DaqD1	3	5	1	4	5	5	4	5	5
FELIX	2?	3	2	3?	3	4	?	3	2
RCE	1	1	5	5	3	3	3	3	2
DaqD2	2	5	5	3	3	3	4	3	2
DaqD3	2	5	5	2	3	3	3	3	2
RCE-upgd	4?	5	5	2	3	3	3	3	2
DaqD4	3	5	5	3	3	4	4	4	2

Table 4: Architecture comparison of factors mentioned in section 1. The rating 1 is the worst and 5 is the best.

The items bellow give some description of columns shown in table 4. The numbers are a very arbitrary scale because although some price estimation work has been done, some of it is still a bit crude. Bigger numbers mean more advantageous.

²This is very rough estimation.

1. Hardware cost estimated per-channels and includes the R&D and prototyping costs. Assuming the same cost for fiber optic cable to the surface for all architectures. The reference is DaqD1 and the estimation is rational relative to cost of DaqD1. The resources FPGA family, electronic components, GPU are selected and estimated regard to minimum functionalities and resources need for front-end DAQ discussed previously.
2. “Filtering high-noise” and “Simplify low-noise” address the high/low noise strategy outlines in section 1. A high number indicates that it is possible to have an extensible system where we could confront an unexpectedly high noise-level with lots of processing and/or not buy and run lots of stuff if the noise turns out to be not so bad.
3. The R&D comprises both development time and cost that includes the cost of engineering-manpower for hardware and firmware, DaqD1 has significantly lower hardware and firmware development effort but it needs CUDA programming that could be done by physicists.
4. The technology life-cycle (TLC) DaqD2 to 4 would have almost the same. the DaqD1 has a very high life time because it uses passive optical components and commercial network components and GPUs which can be replaced easily any time with up to date technology.
5. Fast night repair; human trip to underground for commissioning and maintenance, human trip for shift during night on call-shift to underground.
6. COS; Commercial off shell availability, for DaqD1 almost all components for up-stream DATA to the back-end DAQ could be acquired from off shell commercial components except the FPGA PCI card that is transporting data stream to PC hosted the GPUs. For the other schemes, there is a varying degree of items (networking) that are COS, but some custom FPGA boards.
7. Operating cost; includes power consuming and cooling in the cave, human trips to cavern for maintenance, cost of replacement and spare parts in 20 years (till at least 2044), e.g. DaqD1 has passive optical components till the surface and it is almost zero power consuming, but has a big power-hungry computing farm on the surface with lot of cooling fans on the GPUs that will wear out in 5 years.

NOTE, we do not consider the slow control, clock/timing/command distribution and power supply in this document since they are independent of the up-stream data architecture.

4 Recommendations initial version, need to be discussed with all group

The mentioned architectures in section 3 seem at first look a bit of a radical approach regard to current proposals but they address basics concerns and requirements that are discussed in section1 This will guarantee running the DUNE detector for 20 years with minimum concerns about the maintenance, reliability, components replacement.

- The SOC's are the best candidates especially Xilinx Zynq family is recommended. It has very reach C++ based HLS and as well full solution SDSOC that will reduce firmware R&D hugely even for non-engineer users.
- The architecture DaqD4 is the most optimum solution that relies on off-shell Ethernet components to transfer the data to back-end DAQ PCs and only one (or more per APA) custom design DAQ board based on xx FPGA and XX tool sets. It is midway between the DaqD1 and RCE(DaqD2/3) designs and it addresses the concern indicated at the end of section1's introduction, to provide a 'low-noise hardware trigger' in hardware, that can be used if the detector noise is as good as hoped, but with a fallback Fast Ethernet network powerful enough to run the data

474 into a large computer farm on the surface if noise proves to be a difficulty. This fast network
475 also provides the capability for as large and sophisticated a supernova trigger and buffer as we
476 can afford to be implemented on the surface.

477 **5 Early studies that can be done to evaluate the choices**

- 478 1. The software group are already evaluating the software trigger strategies carefully
- 479 2. We are thinking about the software architecture for some of the specialized trigger data transfers
480 and how these can be implemented in artDAQ and tested at protoDUNE.
- 481 3. What else?

482 **6 Costs comparison on custom and commercial components**

483 Costings for the designs shown here are underway. Currently they are progressed sufficiently to indicate
484 that the designs are competitive at zeroth-order, including a very rough estimate of the development
485 and R&D costs. The costings will soon begin to be refined sufficiently to differentiate between the
486 designs. It is important to refine the estimated cost of R&D effort of each more closely.

Appendix

A Description of the RCE based DAQ

Since, in earlier iterations of this study, we wrote about the standard RCE system, and it is still very relevant to the discussion today, we didn't want to delete it from this note, but we didn't find a way of inserting it into the main flow of the note, so here it is.

The architectural designs of current RCE DAQ systems have two major components in common:

1. Computational Blocks, these provide computational power for data processing and communication between two end-nodes of the DAQ chain, such as DSP, CPU or FPGA. Some blocks also provide slow control and monitoring capabilities.
2. Interface Blocks that facilitate connectivity between Computational blocks with low latency and high-bandwidth for reliable communications between readout electronics and the back-end DAQ.

The DAQ team at SLAC has developed the RCE platform which includes the components of 1 and 2. This DAQ platform is already deployed in the Photon Science experiments at the LINAC Coherent Light Source (LCLS) at SLAC and it is envisioned that it could be used in large-scale systems for future projects such as the upgrade of the ATLAS experiment at the Large Hadron Collider, the Large Synoptic Survey Telescope (LSST) and of course us, the Deep Underground Neutrino Experiment (DUNE).

A.1 RCE Technology

The RCE DAQ architecture is housed in the commercially available Advanced Telecommunications Architecture (ATCA) crate. The key element (Computational block) is based on a Xilinx Zynq System on a Chip (SOC) comprising an FPGA and ARM Linux processor. A group of RCE blocks can ingest and process very high bandwidths of data streams (100s of Gigabits/second). SLAC continues to develop and support new generations of the RCEs with more powerful Zynq SOC's as they become available.

A.1.1 RCE Components

The RCE concept was based on a R&D program for large-scale, high-bandwidth DAQ systems at SLAC. This section is intended to introduce the RCE components and the Cluster On Board (COB) concept. The COB houses several RCEs as shown in Figure 9 below, a description of the components of the Cluster on board (COB) followed by the ATCA crate is given below.

• Cluster on Board (COB)

The COB (Cluster on Board) is an ATCA compliant Front Board serving as the motherboard/carrier for 8 RCEs grouped into 4 Data Processing Modules (DPMs) - each is a computing block. The COB provides the infrastructure for powering with the DPM Interface and for control and monitoring. It hosts the connections to ATCA backplane and user I/O through the P3 connector, front panel displays and Ethernet ports. The embedded Fulcrum network switch ASIC is also directly mounted on the COB PCB.

• Data Processing Module (DPM)

The DPM (Data Processing Module) mezzanine card is the main workhorse of the system. Each DPM hosts two processing RCEs. Each of the processing RCEs is based on a XILINX ZYNQ XC7Z045-2FFG900E with dual Cortex-A9 ARM processors. Up to 12 MGTs can be used for

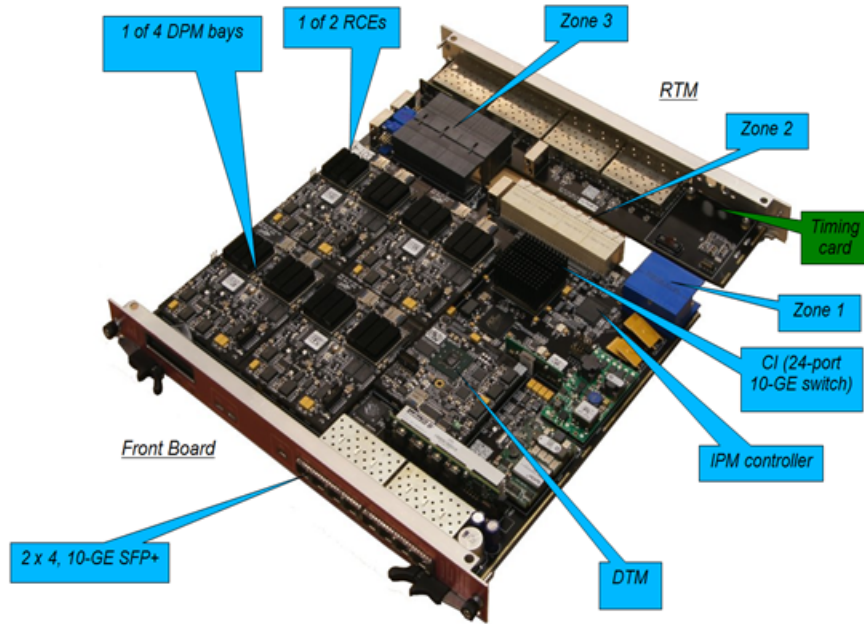


Figure 9: RCE modules on COB and RTM (photo is courtesy of SLAC)

user I/O, however in practice 10Gb/s input per RCE can be expected. There are four DPM bays per COB.

• Data Transportation Module (DTM)

The DTM (Data Transportation Module) is a mezzanine card (one per COB) that manages the network traffic passing through the embedded Fulcrum switch. The DTM core is a single RCE based on ZYNQ XC7Z030-2FBG484E. The DTM is also the TTC interface hub connected to the physical interface on the RTM and responsible for distributing the TTC signal to the DPM RCEs and busy signals back to the TTC system.

• Rear Transition Module (RTM)

The RTM (Rear Transition Module) at the back of the COB connects via the P3 connector is a connectivity board and has almost no active components besides the transceivers interfacing to a variety of I/O possibilities. Two types of standard RTMs have been developed so far: o 16xSFP: with 16 single channel SFP transceiver cages, two channels connected to each RCE. o 96-ch MPO: with 8+8 TX+RX MPO feed-through connectors linked to miniPODs as transceivers. Each MPO is 12-channels and each RCE is connected to a pair of MPOs for TX+RX with 12 channels each. A newer version with 24 channel MPO connectors are also under development. Given the potential variations here they are mostly only relevant for large scale production DAQ systems

• The Cluster-Interconnect (CI)

The 9 RCE (8 DPMs, 1 DTM) nodes on the COB are interconnected locally by the Cluster Interconnect (CI). The CI comprises a low latency, Layer-3 compliant 24-port 10-Gigabit Ethernet switch. The switch is an ASIC based on the Fulcrum Microsystems FM22xx or FM6xxx chip families. To manage the switch, the CI uses the DTM and a protocol that is called the Switch

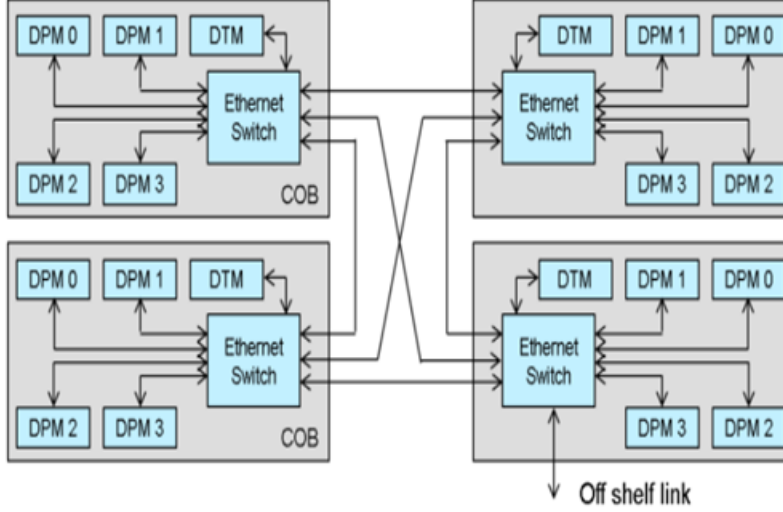


Figure 10: ATCA and COBs Ethernet Clustering, photo is courtesy of SLAC

Management Unit (SMU). The SMU communicates with its corresponding switches through a Fulcrum-defined, private management bus. Software interfaces operating on the SMU allow the switches to be fully configured, monitored and controlled externally. Each of the 8 data processing RCEs has a 10Gbps line connected to the switch while the data transport (DTM) RCE has a 1Gbps connection. Two of the switch ports are routed to the COB front panel, one supporting an SFP+ 10Gbps connection and the other supporting an SFP 1Gbps connection. The remaining Ethernet switch ports are routed to the ATCA backplane. This is shown in Figure 2.

• ATCA infrastructure

The COB containing the RCEs is housed in an ATCA shelf crate with a server computer and a local network setup. The ATCA is built according to a commercial communication and packaging standard, developed and maintained by PCI Industrial Computer Manufacturers Group (PICMG). Features of the ATCA include o A mezzanine standard Advanced Mezzanine Card (AMC) allowing construction of substrate boards.

- Hot-swap capability.
- A Rear-Transition-Module (RTM) allowing for all external cabling to be confined to the backside of the rack enabling the removal of any board without interruption
- Integrated “shelf” support allowing for individual monitoring and control of each board a shelf manager.
- A Low voltage power supply allowing for rack aggregation of power.
- A defined high speed, protocol-agnostic, serial backplane providing point-to-point connections between boards.

As shown in Figure 10, RCEs in one COB can create a cluster in board and share stream of data as well as other COBs in one ATCA.

575 B DUNE Global Trigger, Hardware & Software scheme

576 This part of the appendix also has some slightly older text in it. This is a description of the logical
577 design of the trigger, which is implementable in all the schemes described above. In some of them (e.g.
578 FELIX and DaqD1), all processing steps, including the trigger hit finding are done in software. The
579 current triggers are comprised of following items (at least):

- 580 1. Beam spill
- 581 2. Atmospheric Muon
- 582 3. Supernovae events
- 583 4. Proton decay
- 584 5. Calibration and random triggers

585 The related other projects or work-packages are ;

- 586 • Physics ; trigger requirements
- 587 • DAQ
- 588 • Computation
- 589 • Timing system and RUN control

590 A data processing farm performs a number of processing tasks on both the front-end DAQ and back-
591 end DAQ with time tagged frames to select candidate events and create trigger signal and information.
592 The trigger data flow schemes can be categorized into three distinct classes regarding the hardware
593 and algorithms needed for trigger decision making:

- 594 1. Beam and Calibration and random triggers
- 595 2. Supernovae trigger
- 596 3. Proton decay and Atmospheric Muon triggers

597 For items 2 and 3, we have some idea how to combine them and use the trigger primitive generated in
598 front-end DAQ (in FPGAs) to accelerate the processes and reduce data stream to DTPF bandwidth
599 significantly. Figure 11 shows the general data flow and DAQ components and the probable data flow
600 architecture. In the case of trigger primitive and local ring buffer in front-end DAQ (in FPGAs) one
601 can remove the network B from the architecture. Figure 12 shows mentioned case with more detailed
602 in data processes blocks.

603 The global trigger project steps comprise following items:

- 604 1. Study the physics requirements for current and probable future triggers
- 605 2. Physics and detector electronics simulation
- 606 3. Background and noise estimation including ; ^{39}Ar decay rate, TPC coherent noise and electronics
607 noise.
- 608 4. The data rate, trigger rate, fake event, possible trigger objects and patterns
- 609 5. Algorithm implementation and optimization for running in FPGA node (generating trigger
610 primitive) and/or in trigger farm. Two fundamental trigger primitive algorithms are mentioned
611 below:

- TPC Hit trigger primitives. This would be implemented in FPGA and produce data stream of drift hits on TPC individual channels that would be send out through network to trigger farm for ultimate decision.
 - The “Photon Finder” Algorithm ; The “Photon Finder” (PF) is an algorithm that could be implemented in SSP using signals from SiPM to identify a event in APA. PF will use timing and amplitudes on multiple SiPMs to provide position and timing information for creating trigger decision in DTPF and hardware and software and firmware design for the global trigger.
6. Review and iterate study of feasibility of different schemes between hardware based and software trigger decisions

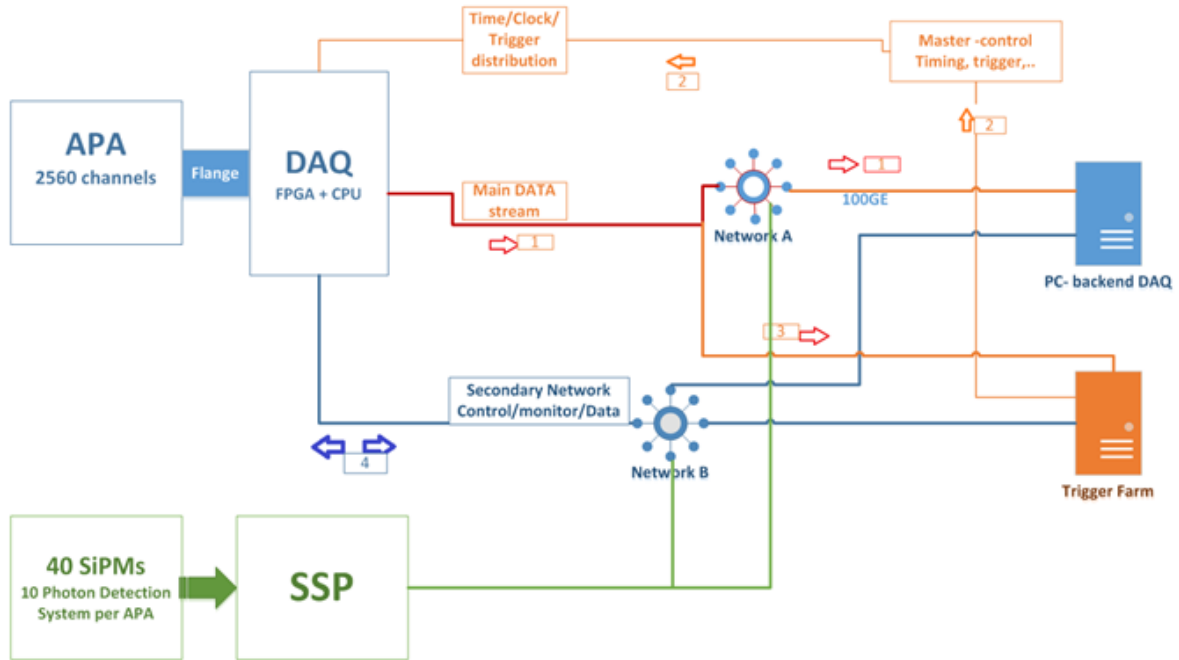


Figure 11: Trigger stream and data flow to trigger farm

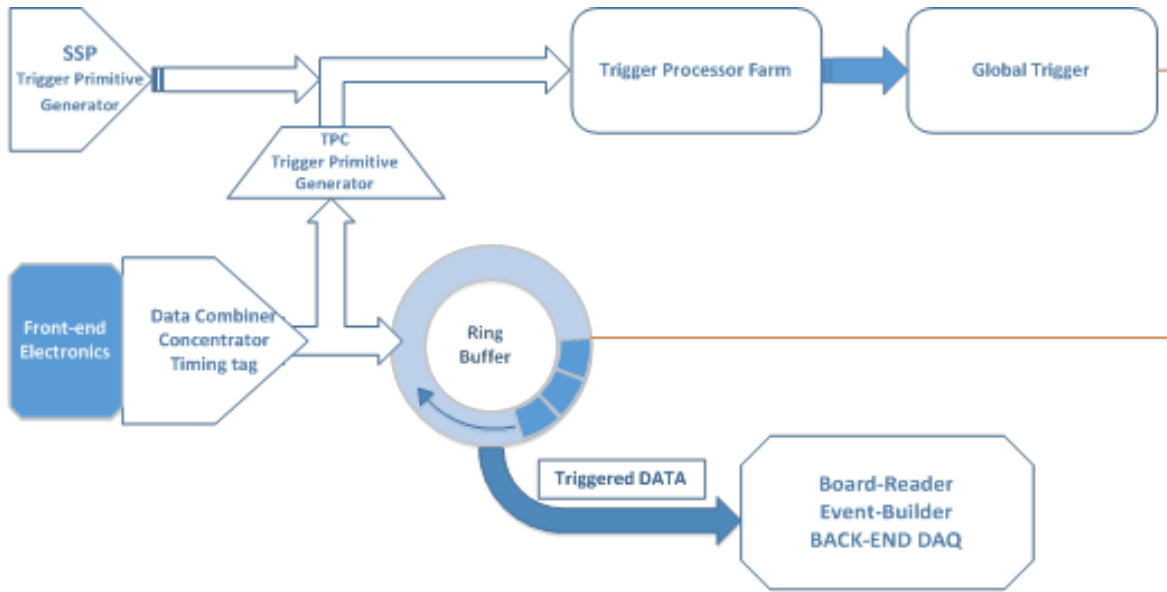


Figure 12: TPC and SSP trigger primitives data flow and ring buffer

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